

National Computing Education Accreditation Council NCEAC

Course Log -Fall 2025

Institution : FAST School of Computing,,National University of Computer & Emerging Sciences, Lahore

Course Name : CS3005-Theory of Automata (BCS-5B)

Week	Date	Duration	Lecture Topics
1	18-Aug-25	1.5	Introduction to TOA, Basics of Discrete Structures, Formal Languages, Union, Intersection, Complement, Kleene Star Kleene Plus
2	25-Aug-25	1.5	Finite Automaton, DFA specifications, DFA Examples
	27-Aug-25	1.5	Transition Table, Extended Transition function, Closure Properties of FA i.e. Union, Intersection
3	1-Sep-25	1.5	Complement and difference of DFAs, State Minimization of DFA
	3-Sep-25	1.5	Extended TF 3rd rule by example, Regular Expressions
4	8-Sep-25	1.5	State Elimination Method
	10-Sep-25	1.5	NFA, ENFA, Kleenes Theorem
5	15-Sep-25	1.5	Kleene's Theorem, Null Closure, ENFA to NFA Conversion, ETF for ENFA
	17-Sep-25	1.5	ETF for NFA, FA Equivalence, Reverse of a language
	20-Sep-25	1.5	Sessional 1
7	29-Sep-25	1.5	Pumping Lemma
	1-Oct-25	1.5	Pumping Lemma
8	6-Oct-25	1.5	Pumping Lemma, CFL and CFG
	8-Oct-25	1.5	Union, Intersection and Kleene Closure of CFLs, Regular Grammar
9	13-Oct-25	1.5	Regular Grammer
	15-Oct-25	1.5	Regular Grammer
	18-Oct-25	1.5	Ambiguous Grammar and solutions
10	20-Oct-25	1.5	PDA, DPDA, NPDA
	22-Oct-25	1.5	PDA Examples
11	27-Oct-25	1.5	CNF, CYK parser
	29-Oct-25	1.5	Mid2 Revision
	1-Nov-25	1.5	Sessional-II
13	10-Nov-25	1.5	Pumping Lemma for CFL
	12-Nov-25	1.5	Properties of CFL and RL, Pumping Lemma examples, introduction to Turing machines
14	17-Nov-25	1.5	Turing Machine as Language acceptors
	19-Nov-25	1.5	Transducers, SUM, Next Binary number, Subtraction
15	24-Nov-25	1.5	Turing machine examples
	26-Nov-25	1.5	Multitape TM
16	1-Dec-25	1.5	Mutitape TM
	3-Dec-25	1.5	Recursive and Recursively Enumerable Languages

Instructor Signature: _____

HOD Signature: _____

Instructor Name: Saba Tariq

HoD Name: Kashif Zafar